



Public Investment under Fiscal Constraints*

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Abstract

The new member states (NMS) of the European Union must comply with the Stability and Growth Pact (SGP) and the investment goals implied by the Lisbon Agenda. However, the SGP rules may result in underinvestment or distortions in the allocation of public expenditure. This paper provides new evidence on the effects of debt sustainability and the SGP fiscal constraints on government expenditure on fixed capital, education and health in OECD countries by estimating government expenditure reaction functions to public debt and cyclical conditions. We find that, at high levels of debt, government capital expenditure and education expenditure are significantly reduced as the debt ratio increases in all OECD countries independently of EMU (or EU) membership. By contrast, neither capital expenditure nor education expenditure is affected by the debt ratio in low-debt countries. These

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findings are robust to the inclusion of the government deficit in the estimated reaction functions. Hence, it appears that EU countries have been constrained in their investment decisions more by the need to ensure debt sustainability than by the rules of the SGP. In low-debt NMS countries, public investment even increases with the debt ratio, a finding that is reassuring for their growth prospects. However, a less optimistic picture emerges when we focus on expenditures on public health and education, as it appears that NMS governments cut such expenditures – even at low levels of debt – as the deficit increases. Problems in controlling total expenditure together with the preventive arm of the SGP may have penalised investment in human capital in the NMS while leaving fixed capital investment unaffected.

I. Introduction

The new member states of the European Union must comply with the Stability and Growth Pact (SGP) and the investment goals implied by the Lisbon Agenda. A trade-off may thus emerge between fiscal targets and economic growth: fiscal constraints may lead to underinvestment or distortions in the allocation of public expenditure and, thus, to lower growth. The aim of this paper is to investigate whether fiscal adjustment needed to ensure debt sustainability or the requirements of the SGP may affect public investment in fixed capital and expenditure on education and health. To assess the impact of the SGP on public investment, the experience of EMU member states is compared with that of other OECD countries.

The constraints implied by debt sustainability or fiscal rules, together with the government's inability to cut consumption or raise taxes, can lead to underinvestment, suboptimal policies and distortions in the allocation of public resources. Fiscal adjustment may hit investment projects that are economically efficient but lack political support and are easier to cut than other expenditures because of weaker opposition from citizens. The SGP, in particular, by imposing limits to the overall budget deficit rather than focusing on debt sustainability,¹ may: (a) lead to lower investment (Balassone and Franco, 2000); (b) favour projects with higher short-term returns over projects with higher returns in the long run (Blanchard and Giavazzi, 2004; Easterly, Irwin and Servén, 2008); and (c) lead to asset decumulation or implicit liabilities (Easterly, 1999) – for example, to inefficient privatisation worsening the fiscal position in the long run (Buiters, 1997; Florio, 2004 and 2007). Advocates of the 'golden rule of public finances', that governments can borrow to fund investment, contend that a limit on the overall government deficit may lead to underinvestment because

¹For details on the SGP, see http://ec.europa.eu/economy_finance/sgp/index_en.htm.

of the stronger incentive to cut capital expenditure relative to other types of expenditures and/or, more generally, to distort the allocation of public resources among alternative programmes. Accordingly, they argue that public investment should be exempted from fiscal constraints and/or that fiscal rules should focus on net debt or even on net worth (for example, Buiter (1984) and Sawyer (2011)).

Balassone and Franco (2000) present descriptive evidence of a potential negative effect of SGP rules on public investment in fixed capital. In particular, focusing on lasting and significant budget consolidation episodes in EU-15 member states between 1980 and 1997, they find that in 25 out of 32 cases, fiscal adjustment was obtained through investment cuts. With some exceptions, the larger reductions occurred in high-debt countries, whereas low-debt countries could increase deficits and investment during cyclical downturns in the 1990s. This opportunity was not available to high-debt countries where fiscal consolidation was still under way and, as a result, the investment to GDP ratio was significantly lower than in low-debt countries.

On the other hand, Galí and Perotti (2003), focusing on EMU countries versus other OECD economies, find that the fall in public investment relative to GDP over the last two decades was no greater in the EU than in other countries. They conclude that the reduction of public investment has been the result of a secular decline affecting all countries at the same time more than the effect of the Maastricht Treaty and the SGP.

We complement and extend previous research in many ways. First, as in Galí and Perotti (2003), we compare public investment in EMU countries with that in other OECD economies, but we explicitly account for debt sustainability motivations by distinguishing between high- and low-debt countries. Indeed, to the extent that debt sustainability constraints are likely to bind only at high levels of debt, their impact should be non-linear, as implicitly recognised by Balassone and Franco (2000). Moreover, we extend the sample of countries to include new member states (NMS) in order to examine the effect of EU membership on their investment decisions. Looking at the NMS separately is important not only because they are a main focus of the analysis, but because they are quite different from the EU-15 countries: they are catching-up economies, with lower public capital stocks and generally lower debt to GDP ratios and higher deficit ratios (see, for example, Orbán and Szapàry (2004)).²

Finally, unlike previous studies, we do not restrict our attention to government capital formation but we also examine government expenditure on education and health. In fact, physical capital accumulation is just one of the many dimensions of public investment; expenditures on education, health, research and technology are surely as important as infrastructure

²Afonso and Alves (2006) argue that in catching-up economies, under certain conditions, running higher deficits can help ensure convergence.

investment as ways of accumulating capital in modern economies. As most public expenditure on education is current expenditure, looking at capital formation alone would underestimate economically relevant public investment.

Although it is widely recognised that government expenditure on education and health plays an important role in human capital accumulation and growth, so far no evidence has been provided on the effect of fiscal constraints on these forms of investment. Likewise, little attention has been paid to the effects of fiscal consolidation on the composition of government expenditure, even though fiscal adjustment may affect growth by changing not only the composition between current and capital expenditure but also the allocation of public funds to different types of expenditures.³ Looking at government expenditure on education and health is a first attempt toward a better understanding of the composition effects of fiscal restraint, which is crucial for a better design of fiscal rules as well as for a better allocation of EU resources among alternative projects.

We provide new evidence on the effects of debt sustainability and SGP fiscal constraints on government expenditure on fixed capital, education and health in OECD countries by estimating government expenditure reaction functions to public debt and cyclical conditions. In particular, we estimate panel regressions with country fixed effects where we distinguish between countries with low and high debt to GDP ratios and between EMU (or EU) member states and other OECD countries to examine whether public investment decisions differ across country groups depending on their fiscal position and membership. Our research strategy is based on the idea that a reduction of public investment (relative to GDP or total government expenditure) should only be observed (or be stronger) at high levels of debt, when debt sustainability is at risk, unless budget deficits are bound by fiscal rules, such as the SGP.⁴ Indeed, absent such rules, only highly indebted countries experiencing debt sustainability problems should reduce public investment as the debt ratio increases. On the other hand, if the SGP were imposing additional constraints on fiscal policies, then the reaction of public investment to the debt and deficit ratios should be different in EMU (or EU) member states from in other OECD economies.

³Composition effects of budget cuts on economic growth were first investigated by Alesina and Perotti (1996). They show that fiscal adjustments occurring with a reduction in public investment are more likely to be soon reversed, while adjustments through cuts in transfers and social security last longer and have a growth-enhancing effect.

⁴A similar approach has been applied by Buti, Franco and Ongena (1997) to study the effect of cyclical downturns on the fiscal policy adopted by EU-15 member states in the presence of the SGP. They show that during periods of severe recession, defined as a fall in GDP by at least 0.75 per cent, countries with higher debt and deficit ratios actually tighten their fiscal policy by reducing their structural primary deficits.

At high levels of debt, we find evidence of a strong negative reaction of both government capital expenditure and education expenditure to the debt ratio. These effects are significant in all highly indebted OECD countries independently of EMU (or EU) membership. By contrast, neither capital expenditure nor education expenditure is affected by the debt ratio in low-debt countries. These findings are robust to the inclusion of the government deficit in the panel regressions.

Hence, it appears that EU countries have been constrained in their investment decisions more by the need to ensure debt sustainability than by the rules of the SGP. At high levels of debt, debt sustainability imposes a constraint on fiscal policy that is strong enough to distort the allocation of government expenditure even in the absence of formal fiscal rules. In particular, public investments in fixed capital and education are penalised relative to other types of expenditures as the debt increases. In light of the European debt crisis, these findings paint a gloomy picture for the growth prospects of highly indebted EU economies.

Finally, in low-debt NMS countries, fixed capital formation also appears not to be affected by the SGP rules. In these countries, public investment even increases with the debt ratio, a finding that is reassuring for their growth prospects. A different picture, however, emerges when we focus on expenditures on public health and education, as it appears that NMS governments cut such expenditures – even at low levels of debt – as the deficit increases. Problems in controlling total expenditure, high deficits and the rules of the preventive arm of the SGP may have penalised investment in human capital in NMS countries while leaving fixed capital investment unaffected.

The paper is organised as follows. The effects of high levels of debt on public investment are investigated in Section II within a simple model of endogenous growth. The estimation results for government capital formation are presented and discussed in Section III. Section IV does the same for government expenditures on education and health. Section V concludes.

II. Public investment and debt sustainability

In this section, we investigate the implications of debt sustainability for the government's choice of public investment. In particular, we examine whether cutting public investment and, thus, the budget deficit may help ensure debt sustainability. In principle, the effect of lower investment on debt sustainability is ambiguous; lower public investment reduces not only the deficit but also the stock of public capital and, thus, output growth and government net wealth, which is arguably a more relevant variable than gross debt for fiscal sustainability.

To address this issue, we rely on a simple one-sector model of endogenous growth where output is produced with both private and public capital, and the government has to service a given stock of public debt. Final goods are produced by households and the government with the following Cobb–Douglas technology with constant returns to scale:

$$(1) \quad Y = K^{1-\alpha} K_G^\alpha,$$

where Y is output and K and K_G denote private and public capital respectively.

The economy's resource constraint is equal to

$$(2) \quad Y = C + G + I_K + I_G,$$

where C and G denote private consumption and government consumption, and I_K and I_G are, respectively, gross private and public investment. The changes in the private and public capital stocks are equal to

$$(3) \quad \dot{K} = I_K - \delta K$$

$$(4) \quad \dot{K}_G = \phi I_G - \delta K_G,$$

where δ is the rate of depreciation, which we assume equal to zero in what follows. Only a fraction, ϕ , of public investment adds to capital accumulation due to leakages in the process of capital formation. Alternatively, $\phi \leq 1$ may capture mismanagement of public funds due to administrative costs, inefficiencies, corruption, etc. To ensure positive growth, we further assume that $\phi > \bar{\phi}$, where $\bar{\phi}$ is determined in what follows.

The assumption of constant returns to scale and the fact that both factors of production can be accumulated yield continuous growth. To derive the rate of growth, suppose that households maximise the following utility function with time preference rate θ :

$$(5) \quad U_0 = \int_0^\infty u(C_t) e^{-\theta t} dt$$

subject to the budget constraint $\dot{W} + C = RW - T$ where $W = K + B$ is private wealth, B is government debt, R is the return on private capital and T is lump-sum taxes net of transfers.

The consumer-producer problem yields the usual solution for the growth rate of consumption. With a constant relative risk aversion (CRRA) utility function, the growth rate is equal to

$$(6) \quad \frac{\dot{C}}{C} = \sigma(R - \theta),$$

where σ is the elasticity of intertemporal substitution (equal to the reciprocal of the coefficient of relative risk aversion). The return on private capital, R , is equal to the marginal product of private capital, i.e. $R = (1-\alpha)(K_G/K)^\alpha$, which is also equal to the return on government debt.

To close the model, consider now the government investment decision. Since taxes are lump sum, the social planner's solution is the same as the decentralised solution because of perfect competition, constant returns to scale and no externalities.⁵ The social planner's maximisation problem yields the following two additional first-order conditions:

$$(7) \quad q\phi = 1$$

$$(8) \quad \dot{q} = Rq - R_G,$$

where q is the shadow price (in terms of consumption goods) of a unit of installed public capital,⁶ and $R = (1-\alpha)(K_G/K)^\alpha$ and $R_G = \alpha(K/K_G)^{1-\alpha}$ denote the marginal product of private and public capital, respectively.

The social planner's solution implies a shadow price, q , for public capital equal to $1/\phi$, i.e. the government must equate the value of an additional unit of public capital with its marginal cost. As the optimal price of capital, $q^* = 1/\phi$, is constant, from equation (8) we have $Rq = R_G$, which, combined with $q^* = 1/\phi$, yields

$$(9) \quad R^* = \phi R_G,$$

which shows that the marginal product of funds invested in public capital must be equal to the marginal product of funds invested in private capital. It follows that the optimal ratio of public to private capital is equal to $K_G^*/K^* = \phi\alpha/(1-\alpha)$ and the optimal growth rate of consumption is equal to

$$(10) \quad \frac{\dot{C}^*}{C^*} = \sigma(R^* - \theta) = \sigma[(1-\alpha)^{1-\alpha}(\phi\alpha)^\alpha - \theta].$$

⁵If a distortionary tax rate on private income, RW , were considered, an externality would be introduced into the model that would lead to lower private capital accumulation than would otherwise be optimal. In this case, a welfare-maximising government would choose public capital so as to raise the ratio of public relative to private capital in order to increase the marginal product of private capital and partly offset the negative effect of the tax rate on consumption growth.

⁶The present-value Hamiltonian is equal to

$$H_t = u(C_t)e^{-\theta t} + \lambda_t(K^{1-\alpha}K_G^\alpha - C_t - G - I_G)e^{-\theta t} + \lambda_t q_t(\phi I_G)e^{-\theta t}.$$

Equation (10) shows that the rate of growth increases with the fraction of investment that adds to public capital, ϕ , which must be $\phi > \bar{\phi} \equiv \theta^{\frac{1}{\alpha}} \left[\alpha(1-\alpha)^{\frac{1-\alpha}{\alpha}} \right]^{-1}$ for growth to be positive.

More generally, when the stock of public capital is not optimal, in a balanced growth path all variables in the economy grow at the same rate $\dot{Y}/Y \equiv g = \sigma(R - \theta)$, which increases with the return on private capital and, thus, with the ratio of public to private capital.

Having described the main features of the model, we now introduce government gross debt, B , to examine the implications of debt sustainability for the government's choice of public investment. The relevant variable for fiscal sustainability is the negative net wealth of the government ('the net debt' for short in what follows), D , which is equal to the difference between the value of gross debt and the value of public capital, i.e. $D = B - qK_G$, where public capital is evaluated at its market price, q .

As gross debt accumulation is equal to $\dot{B} = RB - R_G K_G + I_G + G - T$, the net debt evolves according to⁷

$$(11) \quad \dot{D} = \dot{B} - q\dot{K}_G - \dot{q}K_G = RB - R_G K_G + I_G + G - T - q\dot{K}_G - \dot{q}K_G$$

and the ratio of the net debt to output evolves according to

$$(12) \quad D/Y = (R - g) \frac{D}{Y} + (Rq - R_G) \frac{K_G}{Y} - \left(q - \frac{1}{\phi} \right) \frac{\dot{K}_G}{Y} + \frac{G - T}{Y} - \dot{q} \frac{K_G}{Y},$$

where we have used $\dot{K}_G = \phi I_G$ from equation (4).

In a balanced growth path where all variables grow at the same rate $g > 0$, the ratio of public to private capital is constant, so that the price of public capital, q , is also constant and equal to

$$(13) \quad q = \frac{R_G}{R} = \frac{\alpha}{(1-\alpha)} \frac{K}{K_G}.$$

Hence, the change in the net debt to output ratio is equal to

$$(14) \quad D/Y = (R - g) \frac{D}{Y} - \left(q - \frac{1}{\phi} \right) g \frac{K_G}{Y} + \frac{G - T}{Y}.$$

⁷To cast the analysis in the most favourable conditions for public capital, we assume that the public-capital share of income, $R_G K_G$, is earned by the government – i.e. the return on public capital adds to government revenues – unlike in the growth models stemming from Barro (1990). This is a reasonable hypothesis in the case of physical but not human capital.

In a balanced growth path, fiscal sustainability requires that the net debt ratio does not grow, i.e. $(D/Y) \leq 0$. Thus, if the government chooses the stock of public capital (relative to private capital) optimally, so that $q = 1/\phi$, debt sustainability implies that

$$(15) \quad \frac{T-G}{Y} \geq (R^* - g^*) \left(\frac{B}{Y} - \frac{1}{\phi} \frac{K_G^*}{Y} \right).$$

It follows that any given gross debt ratio can be sustained by a proper choice of government consumption and lump-sum taxes (net of transfers) while setting the public capital to output ratio optimally.

Suppose, however, that the government cannot choose the saving rate $(T-G)/Y$ as in equation (15) – for example, because of political opposition to tax increases and spending cuts. Then, for any given government saving, $\hat{(T-G)}/Y$, there exists a maximum sustainable debt ratio that is defined by the following equation:

$$(16) \quad \frac{B^M}{Y} = \frac{qK_G}{Y} + \frac{g}{(R-g)} \frac{qK_G}{Y} - \frac{g}{\phi(R-g)} \frac{K_G}{Y} + \frac{\hat{T-G}}{(R-g)Y},$$

where the ratio of public to private capital possibly differs from its optimal level and so do the return on private capital, R , and the price of public capital, q .

Equation (16) shows that the maximum sustainable debt ratio depends on the price of public capital, $q = R_G/R$, i.e. on the marginal productivity of public capital relative to the marginal productivity of private capital. It also depends on the ratio of public capital to output both directly and through its effects on the return on debt, R , and the rate of growth, g .

We can now examine whether public investment must be reduced to lower the budget deficit and sustain a higher debt ratio. In principle, the effect of lower public investment and, thus, of a lower stock of public capital is ambiguous. On the one hand, for any given price of public capital, q , a lower stock of public capital (relative to output or private capital) decreases the value of public capital and the rate of growth, impairing debt sustainability. On the other hand, lower public investment reduces the deficit, and a lower stock of public capital raises its marginal productivity (relative to that of private capital) and thus its price, q , while it reduces the return, R , to be paid on debt. These effects allow a higher debt ratio to be sustained.

Assuming a logarithmic utility function, so that $g = R - \theta$, and differentiating B^M/Y in equation (16) with respect to K_G/Y yields⁸

$$(17) \quad \frac{\partial B^M/Y}{\partial K_G/Y} = -\frac{R - \theta + \alpha\theta}{\phi\theta(1 - \alpha)}.$$

This derivative has a negative sign as long as $g = R - \theta > -\alpha\theta$ and, a fortiori, as long as the stock of public relative to private capital is sufficiently high to generate positive growth.

Hence, a lower ratio of public capital to output, which reduces public investment and the government deficit, allows a higher debt ratio to be sustained unless growth becomes negative.⁹ This is because the effect of lower growth due to the reduction in the stock of public capital (relative to output and private capital) is fully offset by an increase in the return on public capital relative to private capital, q , which raises the value of public capital, qK_G , despite the fall in K_G . Intuitively, a higher debt ratio can be sustained by reducing investment expenditure because the return on public capital and its value increase and compensate the negative effect of lower growth.

Hence, at high levels of debt and public capital, a trade-off between debt sustainability and growth emerges as lower public investment decreases the productivity of private capital and growth. Although the price of installed capital, q , is greater than the cost of accumulating it, $1/\phi$, debt sustainability (and the political constraints on raising government saving) prevents the government from undertaking socially desirable investment. This trade-off disappears only when the level of public capital is so low that reducing it further would lead to zero growth.

The analysis suggests that, at high levels of debt, governments facing political opposition to raising taxes or cutting consumption may be forced to reduce public investment to ensure debt sustainability. This could be the case for OECD countries with high debt to GDP ratios, whose behaviour is investigated in the next section.

III. Public capital formation in OECD countries

In this section, we provide evidence on the effect of debt sustainability and the SGP on public investment in fixed capital by estimating public investment reaction functions to government debt for OECD countries over the period 1990–2008.

⁸To obtain equation (17), note that g and q are functions of K_G/Y .

⁹This is purely indicative, since a balanced growth path model requires that growth be strictly positive.

All data refer to general government and are from the OECD National Accounts and the OECD Economic Outlook. We consider two measures of government capital expenditure: (a) gross capital formation and (b) the sum of gross capital formation and acquisitions less disposals of non-financial assets.¹⁰

We estimate a panel regression with country fixed effects and cluster-robust standard errors in which the dependent variable is either the share of government capital expenditure in total expenditure or the ratio of government capital expenditure to GDP. In our baseline specification, government capital expenditure reacts to the ratio of government gross financial liabilities to GDP ('the debt ratio' for short in what follows), the growth rate of GDP and a time trend. All variables are in logs to allow for an easier comparison of the impact of the debt ratio across country groups. All explanatory variables are lagged one period to avoid endogeneity problems.

We use general government gross financial liabilities (from the OECD Outlook database) instead of gross debt because comparable data (across countries) on gross debt from OECD sources are only available for central government.¹¹

As public investment is often used as a countercyclical tool, we include the growth rate of GDP among the explanatory variables to allow government capital expenditure to react to cyclical fluctuations.¹² Finally, we introduce a time trend to control for the alleged secular decline in government fixed capital formation relative to GDP that, Galí and Perotti (2003) contend, has affected public investment in both EU and other OECD economies.

Our research strategy is to distinguish between countries with low and high debt ratios to investigate whether their governments behave differently regarding public investment decisions. The idea is that only highly indebted countries should experience debt sustainability problems and thus reduce government capital expenditure as the debt ratio increases. Therefore, we split the sample into two groups of 15 high-debt countries and 14 low-debt countries depending on their gross financial liabilities (relative to GDP) during the first half of the sample period. The country groups are listed in Table 1. The US is classified as a low-debt country, since its financial

¹⁰Following Galí and Perotti (2003), we exclude in both cases net capital transfers (capital transfers payable less receivable).

¹¹We did not use central government debt because of significant differences in the levels of government across OECD countries. However, we obtained very similar results when, as a robustness check, we estimated the panel using central government debt. It is worth noting that data from other sources, such as the Inter-American Development Bank (IADB), are also for central government debt (see Jaimovich and Panizza (2010)).

¹²It is worth noting that gross capital formation is under the government's control; unlike taxes and social expenditures, it is not a cyclical component of the government budget.

TABLE 1
Classification of countries and estimation samples

		Gross capital formation (GCF)		Expenditure by function		Debt	
		<i>Years</i>	<i>No. of obs.</i>	<i>Years</i>	<i>No. of obs.</i>	<i>Years</i>	<i>No. of obs.</i>
<i>EMU, high debt</i>							
Austria	EU-15	1990–2008	19	1995–2007	13	1990–2008	19
Belgium	EU-15	1990–2008	19	1990–2007	18	1990–2008	19
Finland	EU-15	1990–2008	19	1990–2007	18	1990–2008	19
France	EU-15	1990–2008	19	1995–2007	13	1990–2008	19
Greece	EU-15	2000–2008	9	2000–2007	8	1995–2008	14
Ireland	EU-15	1990–2008	19	1990–2007	18	1990–2008	19
Italy	EU-15	1990–2008	19	1990–2007	18	1990–2008	19
Netherlands	EU-15	1990–2008	19	1995–2007	13	1990–2008	19
Portugal	EU-15	1995–2008	14	1995–2007	13	1995–2008	14
Spain	EU-15	1995–2008	14	1995–2007	13	1990–2008	19
<i>EMU, low debt</i>							
Germany	EU-15	1991–2008	18	1991–2007	17	1991–2008	18
Luxembourg	EU-15	1990–2008	19	1990–2007	18	1990–2008	19
Slovenia	EU	1995–2008	14	2000–2006	7	2001–2007	7
<i>Others, high debt</i>							
Canada		1990–2007	18	1990–2006	17	1990–2008	19
Denmark	EU-15	1990–2008	19	1990–2008	19	1990–2008	19
Hungary	EU	2000–2008	9	2001–2007	7	1995–2008	14
Japan		1996–2007	12	1996–2007	12	1990–2008	19
Sweden	EU-15	1993–2008	16	1995–2007	13	1990–2008	19
<i>Others, low debt</i>							
Australia		1990–2007	18	–	–	1990–2008	19
Czech Republic	EU	1995–2008	14	1995–2007	13	1995–2008	14
Iceland		1998–2008	11	1997–2006	10	1990–2008	19
Korea		1990–2006	17	1995–2006	12	1990–2008	19
New Zealand		1990–2007	18	–	–	1993–2008	16
Norway		1995–2008	14	1996–2007	12	1990–2008	19
Poland	EU	1995–2008	14	2002–2007	6	1995–2008	14
Slovak Republic	EU	1995–2008	14	1995–2007	13	1995–2008	14
Switzerland		1990–2006	17	–	–	1990–2008	19
United Kingdom	EU-15	1990–2008	19	1990–2007	18	1990–2008	19
United States		1990–2008	19	1990–2007	18	1990–2008	19

liabilities are considered net of the holdings of non-marketable government securities by government employee pension funds.¹³ Slovenia, whose data start in 2001, is classified as a low-debt country.

The choice of taking the early years of the sample to classify the countries is consistent with the idea of estimating a government expenditure reaction to an initial debt sustainability problem and presents the further advantage that the chosen partition is remarkably stable until the end of the 1990s. It is, however, worth observing that the country groups would be slightly different if the classification were based on the most recent decade or on the sample averages.¹⁴

In order to control for the effects of EMU or EU membership, and thus the obligations of the SGP, we provide results for two country groupings. In Table 2 we distinguish between EMU member states and other OECD countries, while in Table 3 we consider EU member states versus other OECD countries. Finally, in Table 4 we provide estimates of government gross capital formation for the NMS versus the EU-15 member states. We present results for both EMU and EU countries for two reasons. First, it can be argued that the fiscal constraints imposed by EMU membership are stronger than those associated with EU membership. In particular, EU countries that have not joined the euro, but must nevertheless present ‘Stability and Convergence Programmes’, have lower obligations under the ‘Excessive Deficit Procedure’ and thus the EU is not homogeneous as a group of countries in this respect.¹⁵ The second reason is purely statistical: the distinction between EU and non-EU countries leaves only two countries in the control group of highly indebted non-EU countries – Canada and Japan.

We report the results of panel estimations of gross capital formation with country fixed effects and cluster-robust standard errors in Tables 2 to 4. Table 2 distinguishes between EMU member states and other OECD countries and shows that an increase in the debt ratio reduces gross capital formation in all country groups even if we account for the presence of a time trend. However, the effect is significant only in high-debt countries (independently of their membership), while it is not significant and is weaker in low-debt countries. In highly indebted countries, public investment is significantly reduced both as a percentage of total expenditure (column 1) and relative to GDP (column 2). Interestingly, the elasticity of government capital expenditure to the debt ratio is only slightly higher in EMU member

¹³The OECD overestimates US government financial liabilities because it counts the holdings of non-marketable government securities (7.6 per cent of GDP in 2005) by government employee pension funds as a liability of the government sector on the basis that these schemes are partially funded and, as such, included in the private sector (see OECD 2009 Economic Outlook, annex tables 32 and 33).

¹⁴In particular, Finland and Ireland would be classified as low-debt countries.

¹⁵See Galí and Perotti (2003) for a detailed discussion of this issue.

TABLE 2 Gross capital formation: EMU countries versus other OECD countries

	(1) $GCF/Tot.exp$	(2) GCF/Y	(3) $GCF^*/Tot.exp$	(4) GCF^*/Y	(5) ^b $GCF/Tot.exp$	(6) ^b GCF/Y	(7) $GCF/Tot.exp$	(8) GCF/Y
Debt EMU high _{t-1}	-0.548*** (0.172)	-0.425** (0.158)	-0.590*** (0.180)	-0.467*** (0.168)	-0.539*** (0.195)	-0.447*** (0.156)	-0.559*** (0.184)	-0.469*** (0.152)
Debt EMU low _{t-1}	-0.375 (0.287)	-0.325 (0.313)	-0.444 (0.269)	-0.392 (0.293)	-0.347 (0.327)	-0.296 (0.344)	-0.414 (0.273)	-0.397 (0.306)
Debt others high _{t-1}	-0.490*** (0.134)	-0.393*** (0.131)	-0.548*** (0.124)	-0.450*** (0.114)	-0.496*** (0.142)	-0.421*** (0.133)	-0.501*** (0.134)	-0.497*** (0.118)
Debt others low _{t-1}	-0.120 (0.088)	-0.006 (0.114)	-0.154* (0.086)	-0.042 (0.116)	-0.113 (0.091)	0.007 (0.124)	-0.125 (0.086)	-0.017 (0.110)
GDP growth EMU high _{t-1}	0.030*** (0.007)	0.012* (0.006)	0.029*** (0.009)	0.011 (0.008)	-0.003 (0.007)	-0.014* (0.007)	0.031*** (0.008)	0.016** (0.007)
GDP growth EMU low _{t-1}	0.022** (0.009)	0.013 (0.009)	0.007 (0.015)	-0.001 (0.015)	0.025*** (0.004)	0.018*** (0.004)	0.016* (0.008)	0.008 (0.009)
GDP growth others high _{t-1}	0.010 (0.008)	-0.007 (0.009)	0.022** (0.009)	0.006 (0.009)	-0.002 (0.008)	-0.012 (0.010)	0.012** (0.006)	0.008 (0.005)
GDP growth others low _{t-1}	0.003 (0.007)	-0.006 (0.009)	-0.002 (0.008)	-0.012 (0.009)	0.010 (0.008)	0.009 (0.007)	0.002 (0.007)	-0.006 (0.008)
Deficit EMU high _{t-1}							0.003 (0.007)	0.010 (0.006)
Deficit EMU low _{t-1}							-0.017* (0.009)	-0.014 (0.009)
Deficit others high _{t-1}							0.004 (0.007)	0.017** (0.007)
Deficit others low _{t-1}							-0.004 (0.007)	0.001 (0.007)
Trend	-0.007 (0.005)	-0.013** (0.005)	-0.008 (0.006)	-0.015** (0.006)	-0.006 (0.006)	-0.013** (0.006)	-0.006 (0.006)	-0.011* (0.006)
Constant	3.572*** (0.425)	2.619*** (0.432)	3.831*** (0.461)	2.880*** (0.472)	3.578*** (0.477)	2.663*** (0.450)	3.595*** (0.432)	2.670*** (0.427)
No. of observations	450	447	436	433	450	447	442	439
R ²	0.316	0.267	0.304	0.262	0.283	0.266	0.317	0.277
No. of countries	29	29	28	28	29	29	29	29

Notes to Table 2

^aIn columns 3 and 4, 'gross capital formation plus acquisitions less disposals' is used as the dependent variable instead of 'gross capital formation'.

^bIn columns 5 and 6, 'HP output gap' is used as an explanatory variable instead of 'GDP growth rate'.

Note: Robust standard errors are given in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

states than in the other highly indebted countries, followed by low-debt EMU countries and, finally, by the other low-debt countries. Hence, a strong and significant negative reaction of public investment to the debt ratio emerges at high levels of debt independently of EMU membership, a result that suggests the irrelevance of the formal constraints imposed by the SGP.

A similar picture emerges when we consider, in columns 3 and 4, as a dependent variable, gross capital formation plus acquisitions less disposals of non-financial assets as a percentage of total expenditure and relative to GDP, respectively. Indeed, both the size and the significance of the coefficients on the debt ratio do not change much when we focus on gross capital formation plus acquisitions less disposals. This suggests that the observed negative impact of the debt ratio on public investment was not brought about by the wave of privatisations that characterised European countries over the past three decades.

Finally, we observe that both definitions of government capital expenditure in terms of GDP show a significant negative time trend. In the case of high-debt EMU member states, GDP growth also appears to positively affect government capital expenditure (excluding acquisitions less disposals) both as a percentage of total expenditure and relative to GDP. The latter result confirms the finding by Galí and Perotti (2003) of a greater (though declining) pro-cyclicality of fiscal stance in EMU countries. To check whether our estimates are robust to a different specification of the cycle, we use a Hodrick–Prescott (HP) filter to decompose each country's GDP into the output gap and the trend GDP and we then re-estimate the panels in columns 1 and 2 using the output gap in place of GDP growth and normalising government capital expenditure with trend GDP. As shown in columns 5 and 6, when cyclical conditions are measured by the output gap, there is no evidence of fiscal pro-cyclicality in highly indebted EMU countries; if anything, the negative coefficient on the output gap (though significant only at the 10 per cent level) is indicative of a countercyclical reaction of government capital expenditure. By contrast, a significant positive reaction to the output gap now emerges in the case of low-debt EMU member states. It is, however, worth noting that the elasticities of capital expenditure to the debt ratio are remarkably robust to the change in specification.

These findings suggest that the SGP does not set additional constraints on the fiscal policies of EMU member states, in that, at high levels of debt, government capital expenditure is significantly reduced as the debt to GDP

ratio increases, in all OECD countries independently of EMU membership. Our findings are consistent with the evidence provided by Galí and Perotti (2003) but qualify their results in an important way: the decline of public investment observed in all OECD countries is due not just to historical factors but to a common concern for debt sustainability. In other words, the common behaviour of the two country groups at high levels of debt does not allow us to conclude that the fiscal rules of the SGP are not binding but that, absent such rules, debt sustainability imposes a constraint on fiscal policy that is strong enough to equally distort the allocation of government expenditure.

However, since the SGP imposes a stronger constraint on the deficit than on the debt ratio, an obvious objection to our analysis is that the deficit to GDP ratio should be included in the estimated reaction function. Panel estimates controlling for the deficit ratio are presented in the last two columns of Table 2.¹⁶ These estimates strengthen our previous conclusions: the elasticity of gross capital formation to the debt ratio increases in highly indebted countries outside the euro area and becomes very similar to that of EMU member states. Indeed, at high levels of debt, higher deficits do not place any additional constraint on government capital spending.

There is, however, some evidence of a negative effect of the deficit on the share of capital expenditure in low-debt EMU countries. Although this finding suggests that the 3 per cent deficit limit could be binding at low levels of debt, the effect is significant only at the 10 per cent level.

Table 3 reports results for the alternative country grouping that distinguishes between EU and other OECD countries in order to examine whether the fiscal rules of the preventive arm of the SGP bind the investment decisions of all EU member states. Specifically, we want to check whether the negative effect of high levels of debt on capital expenditure outside the euro area emerges because of a common concern for debt sustainability or just because of the inclusion within this group of Denmark, Hungary and Sweden, which are subject to (milder) SGP fiscal rules.

The result of this exercise is striking: not only does the reaction of government capital expenditure to the debt ratio in highly indebted non-EU countries remain significant at the 1 per cent level but also its elasticity increases above that of highly indebted EU countries. This can easily be seen by comparing the coefficients on the debt ratios in Tables 2 and 3 for the

Notes to Table 3

^aIn columns 3 and 4, 'gross capital formation plus acquisitions less disposals' is used as the dependent variable instead of 'gross capital formation'.

^bIn columns 5 and 6, 'HP output gap' is used as an explanatory variable instead of 'GDP growth rate'.

Note: Robust standard errors are given in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

¹⁶The deficit is measured as net borrowing (from OECD National Accounts), which is the relevant definition for the SGP. It is the only variable not in logs.

TABLE 3 Gross capital formation: EU countries versus other OECD countries

	(1) $GCF/Tot.exp$	(2) GCF/Y	(3) $GCF^*/Tot.exp$	(4) GCF^*/Y	(5) ^b $GCF/Tot.exp$	(6) ^b GCF/Y	(7) $GCF/Tot.exp$	(8) GCF/Y
Debt EU high _{t-1}	-0.514*** (0.159)	-0.395** (0.144)	-0.565*** (0.164)	-0.446*** (0.150)	-0.507*** (0.179)	-0.417*** (0.144)	-0.534*** (0.178)	-0.450*** (0.147)
Debt EU low _{t-1}	-0.099 (0.190)	-0.015 (0.219)	-0.180 (0.196)	-0.100 (0.229)	-0.039 (0.222)	0.031 (0.251)	-0.091 (0.194)	-0.031 (0.231)
Debt others high _{t-1}	-0.803*** (0.070)	-0.684*** (0.021)	-0.793*** (0.130)	-0.673*** (0.067)	-0.818*** (0.039)	-0.726*** (0.027)	-0.784*** (0.054)	-0.767*** (0.055)
Debt others low _{t-1}	-0.171* (0.087)	-0.054 (0.131)	-0.188** (0.088)	-0.071 (0.135)	-0.163* (0.088)	-0.037 (0.146)	-0.170** (0.077)	-0.055 (0.118)
GDP growth EU high _{t-1}	0.027*** (0.007)	0.009 (0.006)	0.029*** (0.008)	0.012* (0.006)	-0.001 (0.007)	-0.013** (0.006)	0.029*** (0.008)	0.015** (0.006)
GDP growth EU low _{t-1}	0.018 (0.011)	0.007 (0.014)	0.009 (0.014)	-0.003 (0.017)	0.023** (0.009)	0.017 (0.013)	0.021** (0.010)	0.012 (0.012)
GDP growth others high _{t-1}	0.010 (0.016)	-0.010 (0.011)	0.012 (0.010)	-0.008 (0.006)	-0.010 (0.012)	-0.014*** (0.005)	0.003 (0.009)	0.001 (0.009)
GDP growth others low _{t-1}	-0.004 (0.004)	-0.011** (0.004)	-0.006 (0.004)	-0.014*** (0.004)	0.010 (0.009)	0.010 (0.006)	-0.006 (0.004)	-0.014** (0.006)
Deficit EU high _{t-1}							0.003 (0.006)	0.011* (0.006)
Deficit EU low _{t-1}							0.007 (0.009)	0.014 (0.011)
Deficit others high _{t-1}							-0.006 (0.009)	0.010 (0.008)
Deficit others low _{t-1}							-0.014** (0.006)	-0.012 (0.008)
Trend	-0.007 (0.005)	-0.014** (0.005)	-0.009* (0.006)	-0.015** (0.006)	-0.007 (0.005)	-0.013** (0.006)	-0.008 (0.006)	-0.012* (0.006)
Constant	3.608*** (0.440)	2.644*** (0.443)	3.859*** (0.477)	2.893*** (0.483)	3.566*** (0.493)	2.656*** (0.462)	3.635*** (0.464)	2.685*** (0.454)
No. of observations	450	447	436	433	450	447	442	439
R ²	0.318	0.264	0.304	0.259	0.285	0.264	0.327	0.284
No. of countries	29	29	28	28	29	29	29	29

control group of highly indebted other OECD countries. Moreover, if we compare the impact of the debt ratio in highly indebted EMU and EU countries, we observe coefficients of approximately the same size, a further indication that debt sustainability, as opposed to the different fiscal rules applying to EU member states, is the key constraint binding investment decisions. These results hold independently of whether we focus on the share of government capital expenditure in total expenditure or on its ratio to GDP and are also valid for the definition of capital formation including acquisitions less disposals of financial assets (columns 3 and 4). Furthermore, the estimates are robust to the use of the output gap (columns 5 and 6) and, more importantly, to the inclusion of the deficit to GDP ratio (columns 7 and 8).¹⁷

Hence, a main finding of the analysis is that the behaviour of the control group of highly indebted countries does not change independently of whether this group includes or excludes EU countries that do not adopt the euro. At high levels of debt, governments cut public investment, as the debt increases, independently of whether they have to abide by the rules of the SGP.

A second finding is that EMU membership seems to matter at low levels of debt: in Germany, Luxembourg and Slovenia (low-debt EMU countries), public investment decreases, though not significantly, with the debt ratio, whereas it shows no reaction when all low-debt EU countries are considered together. Indeed, by comparing Tables 2 and 3, it can be seen that the coefficient on the debt ratio falls substantially when all low-debt EU countries are considered together.

A tentative explanation for this evidence is that the Excessive Deficit Procedure (the dissuasive arm of the SGP) may have resulted in a stronger constraint on the fiscal policy of EMU members, for instance forcing cuts in deficits and thus in public investment even at low levels of debt.¹⁸ As the group of EU countries with a low debt ratio not adopting the euro mainly includes new member states, such as the Czech Republic, Poland and the Slovak Republic, besides the UK, it is interesting to further examine their behaviour and compare it with that of the group of EU-15 countries.

Table 4 reports estimates of government gross capital formation distinguishing between highly indebted EU-15 countries, low-indebted EU-15 countries and the NMS with a low debt ratio (thus excluding Hungary). The results in columns 1 to 8 are for the same specifications as presented in

¹⁷Interestingly, when the output gap is used in place of GDP growth in order to capture a possible reaction to cyclical downturns, a significant countercyclical reaction of capital expenditure (relative to trend GDP) is detected for both groups of highly indebted countries (see column 6).

¹⁸It is, however, worth noting that no reaction of capital expenditure to the deficit is observed for low-debt EU member states (see columns 7 and 8 of Table 3).

TABLE 4 Gross capital formation: EU-15 countries versus new member states

	(1) $GCF/Tot.exp$	(2) GCF/Y	(3) $GCF^*/Tot.exp$	(4) GCF^*/Y	(5) ^b $GCF/Tot.exp$	(6) ^b GCF/Y	(7) $GCF/Tot.exp$	(8) GCF/Y
Debt EU-15 high _{t-1}	-0.527*** (0.177)	-0.414** (0.167)	-0.584*** (0.188)	-0.472** (0.181)	-0.543*** (0.178)	-0.428** (0.169)	-0.532** (0.186)	-0.444** (0.160)
Debt EU-15 low _{t-1}	-0.410 (0.277)	-0.358 (0.299)	-0.430 (0.268)	-0.380 (0.286)	-0.412 (0.288)	-0.361 (0.312)	-0.422 (0.282)	-0.394 (0.309)
Debt NMS low _{t-1}	0.201*** (0.067)	0.339*** (0.066)	0.103 (0.068)	0.236*** (0.071)	0.211*** (0.068)	0.356*** (0.068)	0.193** (0.070)	0.317*** (0.075)
GDP growth EU-15 high _{t-1}	0.027*** (0.007)	0.010 (0.006)	0.030*** (0.008)	0.013* (0.007)	0.029*** (0.007)	0.011* (0.006)	0.028*** (0.008)	0.013* (0.006)
GDP growth EU-15 low _{t-1}	0.008 (0.008)	-0.004 (0.011)	-0.005 (0.011)	-0.017 (0.012)	0.009 (0.009)	-0.002 (0.011)	0.007 (0.008)	-0.002 (0.009)
GDP growth NMS low _{t-1}	0.028*** (0.007)	0.019* (0.010)	0.026** (0.012)	0.014 (0.015)	0.027*** (0.007)	0.018* (0.009)	0.033*** (0.007)	0.026*** (0.005)
Deficit EU-15 high _{t-1}							0.002 (0.007)	0.007 (0.006)
Deficit EU-15 low _{t-1}							-0.001 (0.010)	0.004 (0.011)
Deficit NMS low _{t-1}							0.015 (0.016)	0.021 (0.018)
Trend	-0.010 (0.007)	-0.017** (0.007)	-0.012 (0.009)	-0.020** (0.009)	-0.010 (0.007)	-0.018** (0.008)	-0.009 (0.008)	-0.015* (0.008)
Constant	3.690*** (0.645)	2.869*** (0.640)	4.077*** (0.736)	3.266*** (0.738)	3.737*** (0.651)	2.905*** (0.648)	3.665*** (0.650)	2.851*** (0.623)
No. of observations	303	301	289	287	303	301	299	297
R ²	0.344	0.329	0.323	0.311	0.350	0.334	0.348	0.341
No. of countries	19	19	18	18	19	19	19	19

^aIn columns 3 and 4, 'gross capital formation plus acquisitions less disposals' is used as the dependent variable instead of 'gross capital formation'.

^bIn columns 5 and 6, 'HP output gap' is used as an explanatory variable instead of 'GDP growth rate'.

Note: Robust standard errors are given in parentheses. *** p < 0.01; ** p < 0.05; * p < 0.1.

previous tables. The reaction of government capital expenditure (relative to both total expenditure and GDP) to the debt ratio is negative and significant in highly indebted EU countries; it is negative but not significant in low-indebted EU-15 countries; and it is positive and highly significant for the low-indebted NMS group (see columns 1 and 2). These results hold, with just one exception, for all specifications considered and, in particular, when we control for the deficit; the latter does not affect the capital expenditure of any group of countries. The exception is that the positive reaction to the debt ratio shown by the NMS is no longer significant when we focus, in column 3, on the share of gross capital formation plus acquisitions less disposals of non-financial assets (in total expenditure); this result is easily explained by the large privatisation programmes in these countries over the period considered.

Although this evidence does not allow us to identify the relative roles of EU and EMU membership, it clearly points to the negative effect of government debt on investment decisions at high levels of debt when fiscal sustainability is an issue. While public investment has decreased almost everywhere over the past three decades, this decline has been stronger when associated with high and rising debt levels. By contrast, in low-debt countries, the debt ratio does not appear to significantly affect public investment. More importantly, in the case of the NMS, an increase in the debt ratio even favours capital formation. Indeed, there is strong evidence that NMS countries have not been constrained, so far, in their investment decisions by EU accession and the rules set by the SGP. This finding is reassuring for the growth prospects and potential of these countries.

IV. Public expenditures on education and health in OECD countries

Public investment in fixed capital does not exhaust the possible channels through which the government can stimulate economic growth. Modern growth theory emphasises the importance of human capital for growth: knowledge is considered a public good generating positive externalities on productivity.

Lucas (1988) argues that investment in education increases GDP due to human capital externalities. The latter arise if the presence of educated workers makes other workers more productive, creating learning opportunities through social and professional interactions. Another channel is the facilitation of adoption and creation of new ideas and technologies. Finally, externalities can be generated by a reduction of crime, improved health and higher tax revenues (McMahon, 1999).

The benefits of human capital are the main rationale for government involvement in education (Hanushek, 2002). As human capital generates

positive externalities, mainly due to the link with technological change and to the indirect positive effect of education on productivity, the social rate of return is larger than the private one. This implies that the evaluation of expenditure on education programmes should take into account these spillovers in order to obtain reliable estimates of the effects of government policies.

Despite the importance of the contribution of human capital, uncertainties remain about the magnitude of its social return because of various econometric and measurement problems (Benhabib and Spiegel, 1994; Bassanini and Scarpetta, 2001). Taking into account positive externalities, some studies show that the return on investment in human capital can be higher than that on investment in physical capital. For EU countries, de la Fuente and Ciccone (2003) report an estimated direct return on schooling expenditure comparable to that of physical capital; they claim that, once non-market returns to education and benefits for social cohesion are considered, the returns may be even higher. For Spain, de la Fuente, Domenech and Jimeno (2003) show that the social return on human capital is higher than the return on investment in physical capital.

Another type of government expenditure that contributes to human capital formation is health expenditure. While such expenditure is often neglected, empirical evidence shows that it has a positive and significant effect on aggregate output (Rivera and Currais, 1999; Bloom, Canning and Sevilla, 2004). Moreover, the interactions between the different types of expenditure, such as education and health, reinforce mutual spillovers and growth (Agenor and Neanidis, 2006).

Therefore, government expenditure on education and health, though mostly accounted as current expenditure, is another important form of public investment that deserves special attention.

In this section, we examine the effects of debt sustainability and SGP rules on government expenditure on education and health by relying on the OECD classification of general government expenditure by type of function, which is available for the period 1990–2007 (see Table 1). We repeat the same panel estimation with fixed effects and cluster-corrected standard errors as described in the previous section using the expenditures on education and on public health as dependent variables. As before, we consider both the share of government expenditure on education (or health) in total expenditure and the ratio of government expenditure to GDP and estimate their reaction to the debt ratio and the growth rate of GDP (both lagged by one year).¹⁹

In a first panel, we distinguish four groups of countries: highly indebted EMU members; low-indebted EMU members; highly indebted other OECD countries; and low-indebted other OECD countries. We do not present

¹⁹We do not present estimates with a time trend as the latter turned out to be not significant.

estimates for the alternative classification based on EU membership as they do not add information beyond that obtained by considering EMU membership.

The reaction of government expenditure on education to the debt ratio and GDP growth is reported in Table 5. Column 1 shows that an increase in the debt ratio significantly reduces the share of education expenditure in highly indebted countries independently of EMU membership, whereas it has no effect in low-indebted countries. Quite surprisingly, the effect is stronger for countries outside the euro area; their share of education expenditure exhibits an elasticity to the debt ratio that is almost twice that for highly indebted EMU members. These results are robust to the use of the output gap in place of GDP growth (column 3) and to the inclusion of the deficit except for the magnitude of the elasticity (column 5). Although the deficit displays a significant negative effect on the share of education expenditure in highly indebted EMU member states, an even stronger effect emerges in other OECD economies, independently of their level of indebtedness. Moreover, the deficit appears to have a positive impact on the ratio of education expenditure to GDP in EMU countries and no significant effect in other OECD countries (see column 6), a fact that is difficult to reconcile with the relevance of the SGP deficit limit.

Hence, there is strong evidence that debt sustainability induces a composition effect in government expenditure. At high levels of debt, investment in education is penalised relative to other types of expenditure when the debt increases. Once again, this effect appears to be unrelated to the formal constraints imposed by the SGP in that EMU membership plays no role; the effect of the debt ratio is weaker for EMU members than for other OECD countries.

On the other hand, when we focus on the ratio of education expenditure to GDP (columns 2 and 4), the debt ratio is no longer significant for any group of countries. Only if we control for the deficit ratio does the coefficient on the debt ratio become significant, though only at the 10 per cent level, for highly indebted countries outside the euro area (column 6). This mixed evidence could be explained by a positive co-movement between total government expenditure and public debt that leaves education expenditure unaffected. The estimates in column 8 show that, once we control for total expenditure, an increase in the debt ratio still negatively affects education expenditure to GDP significantly in highly indebted

Notes to Table 5

^aIn columns 3 and 4, 'HP output gap' is used as an explanatory variable instead of 'GDP growth rate'.

^bIn columns 7 and 8, 'Total government expenditure to GDP ratio' is used as an explanatory variable instead of 'Deficit to GDP ratio'.

Note: Robust standard errors are given in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

TABLE 5 Government expenditure on education: EMU countries versus other OECD countries

	(1) <i>Edu/Tot.exp</i>	(2) <i>Edu/Y</i>	(3) ^a <i>Edu/Tot.exp</i>	(4) ^a <i>Edu/Y</i>	(5) <i>Edu/Tot.exp</i>	(6) <i>Edu/Y</i>	(7) ^b <i>Edu/Tot.exp</i>	(8) ^b <i>Edu/Y</i>
Debt EMU high _{t-1}	-0.085*** (0.013)	0.044 (0.055)	-0.074*** (0.022)	0.022 (0.072)	-0.061** (0.022)	-0.003 (0.066)	-0.027 (0.024)	-0.090*** (0.032)
Debt EMU low _{t-1}	0.018 (0.036)	-0.044 (0.082)	0.023 (0.043)	-0.065 (0.079)	0.023 (0.038)	-0.038 (0.084)	0.015 (0.030)	-0.055 (0.064)
Debt others high _{t-1}	-0.152*** (0.049)	-0.009 (0.041)	-0.162*** (0.048)	-0.031 (0.030)	-0.089*** (0.026)	-0.078* (0.041)	-0.108** (0.043)	-0.124** (0.047)
Debt others low _{t-1}	-0.010 (0.031)	0.070 (0.060)	-0.027 (0.055)	0.069 (0.070)	-0.005 (0.036)	0.084 (0.053)	0.066 (0.049)	0.061 (0.054)
GDP growth EMU high _{t-1}	0.003 (0.003)	-0.017*** (0.003)	0.005 (0.005)	0.001 (0.004)	0.001 (0.002)	-0.013*** (0.003)	-0.001 (0.002)	-0.007** (0.003)
GDP growth EMU low _{t-1}	0.004*** (0.001)	-0.005*** (0.000)	0.003*** (0.001)	-0.004*** (0.000)	0.005*** (0.001)	-0.004*** (0.001)	0.005*** (0.001)	-0.001* (0.000)
GDP growth others high _{t-1}	0.000 (0.001)	-0.016* (0.008)	-0.007** (0.003)	-0.010*** (0.003)	-0.007** (0.003)	-0.010** (0.004)	-0.004 (0.003)	-0.008* (0.004)
GDP growth others low _{t-1}	0.012 (0.008)	0.004 (0.005)	-0.006 (0.008)	0.003 (0.005)	0.010 (0.007)	0.005 (0.005)	0.007 (0.006)	0.006 (0.004)
Deficit EMU high _{t-1}					-0.005** (0.002)	0.009*** (0.003)	-0.285*** (0.090)	0.646*** (0.098)
Deficit EMU low _{t-1}					0.003*** (0.001)	0.004*** (0.000)	0.193*** (0.044)	0.586*** (0.074)
Deficit others high _{t-1}					-0.007* (0.004)	0.009 (0.007)	-0.208 (0.204)	0.629* (0.313)
Deficit others low _{t-1}					-0.009* (0.005)	0.001 (0.005)	-0.648** (0.307)	0.206 (0.260)
Constant	2.765*** (0.062)	1.595*** (0.121)	2.788*** (0.081)	1.634*** (0.151)	2.683*** (0.062)	1.698*** (0.132)	3.765*** (0.343)	-0.038 (0.360)
No. of observations	346	347	347	348	341	341	341	341
R ²	0.142	0.143	0.093	0.032	0.209	0.237	0.283	0.329
No. of countries	26	26	26	26	26	26	26	26

TABLE 6 Government expenditure on education: EU-15 countries versus new member states

	(1) $Edu/Tot.exp$	(2) Edu/Y	(3) ^a $Edu/Tot.exp$	(4) ^a Edu/Y	(5) $Edu/Tot.exp$	(6) Edu/Y	(7) ^b $Edu/Tot.exp$	(8) ^b Edu/Y
Debt EU-15 high _{t-1}	-0.109*** (0.026)	0.031 (0.050)	-0.101*** (0.031)	0.013 (0.062)	-0.066** (0.028)	-0.008 (0.064)	-0.036 (0.032)	-0.083** (0.033)
Debt EU-15 low _{t-1}	0.023 (0.024)	-0.041 (0.063)	0.050 (0.035)	-0.056 (0.059)	0.022 (0.023)	-0.042 (0.063)	0.027 (0.029)	-0.055 (0.053)
Debt NMS low _{t-1}	0.038* (0.019)	0.118** (0.051)	0.017 (0.191)	0.136 (0.143)	0.085*** (0.008)	0.125*** (0.038)	0.153*** (0.041)	0.140** (0.054)
GDP growth EU-15 high _{t-1}	0.003 (0.003)	-0.015*** (0.003)	0.004 (0.003)	-0.000 (0.004)	-0.001 (0.002)	-0.013*** (0.003)	-0.002 (0.002)	-0.008*** (0.002)
GDP growth EU-15 low _{t-1}	0.008* (0.004)	-0.002 (0.003)	0.007 (0.004)	-0.004*** (0.001)	0.005*** (0.002)	-0.003 (0.003)	0.006*** (0.002)	0.001 (0.003)
GDP growth NMS low _{t-1}	0.030* (0.015)	0.017 (0.010)	-0.029 (0.033)	-0.001 (0.015)	0.023* (0.012)	0.016* (0.009)	0.018 (0.011)	0.015 (0.009)
Deficit EU-15 high _{t-1}					-0.007*** (0.002)	0.006* (0.003)	-0.354*** (0.101)	0.556*** (0.116)
Deficit EU-15 low _{t-1}					-0.008 (0.007)	-0.002 (0.004)	-0.145 (0.237)	0.451*** (0.095)
Deficit NMS low _{t-1}					-0.017*** (0.005)	-0.002 (0.005)	-1.136*** (0.038)	-0.218*** (0.034)
Constant	2.681*** (0.079)	1.575*** (0.146)	2.674*** (0.128)	1.604*** (0.195)	2.570*** (0.080)	1.674*** (0.180)	4.031*** (0.292)	0.209 (0.310)
No. of observations	260	260	261	261	257	257	257	257
R ²	0.211	0.191	0.126	0.041	0.295	0.243	0.427	0.337
No. of countries	19	19	19	19	19	19	19	19

^aIn columns 3 and 4, 'HP output gap' is used as an explanatory variable instead of 'GDP growth rate'.

^bIn columns 7 and 8, 'Total government expenditure to GDP ratio' is used as an explanatory variable instead of 'Deficit to GDP ratio'.

Note: Robust standard errors are given in parentheses. *** p < 0.01; ** p < 0.05; * p < 0.1.

countries. As total expenditure and public debt rise, investment in education is reduced relative to other types of expenditures.²⁰

This evidence leads us to the conclusion that debt sustainability concerns have forced highly indebted countries to cut spending on public education relative to other types of expenditure, while the formal rules of the SGP have played little or no role.

Next, we restrict our attention to EU member states to investigate whether the distribution of government expenditure in the NMS has been different from that in the EU-15. To this end, we distinguish three groups of countries: high-debt EU-15 countries; low-debt EU-15 countries; and low-debt NMS countries (thus excluding Hungary). The results of the estimation, presented in Table 6, show that, in highly indebted EU-15 countries, the impact of the debt ratio on the share of education expenditure (in total expenditure) is negative and significant in baseline specifications (columns 1 and 3). By contrast, in low-indebted NMS countries, higher debt ratios have a positive effect on the share of education expenditure that is significant in all specifications except when GDP growth is replaced by the output gap (column 3). Finally, it is worth noting that the debt ratio has no effect on the expenditure decisions of low-debt EU-15 countries. This evidence is consistent with that already observed for these three groups of countries in the case of gross capital formation. On the other hand, the ratio of education expenditure to GDP shows no reaction to the debt ratio in all EU-15 members, independently of their level of debt (columns 2 and 4), while it remains significant at the 5 per cent level in NMS countries in the baseline specification (column 2).

Hence, there is strong evidence that high and rising debt levels significantly reduce spending on education relative to other types of expenditure in highly indebted European countries, whereas education expenditure is not adversely affected at low or moderate levels of debt. In fact, in low-debt NMS countries, education expenditure is even favoured by the accumulation of public debt.

To further investigate whether the rules set by the SGP have imposed a constraint on fiscal policy in EU member states, we add the deficit to GDP ratio to the explanatory variables of the panel regression. The estimates reported in columns 5 and 6 suggest that the effects of the debt ratio are robust to the introduction of the deficit ratio. However, the deficit has an impact on the expenditure composition; the effect is negative and significant in highly indebted EU-15 countries and in low-indebted NMS economies, while it is not significant in low-indebted EU-15 countries. As this evidence may simply reflect a greater stability of education expenditure relative to total expenditure, it is worth examining the relation between education expenditure and total expenditure (both relative to GDP), which is presented

²⁰The elasticity of education expenditure to total expenditure is less than 1 (see column 8).

in column 8. While in EU-15 countries the elasticity of education expenditure to total expenditure is positive (though less than 1), in NMS countries this elasticity is negative and strongly significant. It appears that NMS governments have cut expenditure on education as total expenditure and the deficit increased, even at low levels of debt. Hence, there is some evidence that the inability to control government expenditure, together with EU accession and the rules of the preventive arm of the SGP, may have distorted expenditure allocation in NMS countries.

Finally, we examine government expenditure on public health, as health is also an important factor in human capital formation. We proceed as in the case of government expenditure on education, with results reported in Tables 7 and 8.

We find, contrary to our earlier results on gross capital formation and education expenditure, that an increase in government debt reduces government expenditure on public health only in EMU member states and this effect is significant independently of whether GDP growth or the output gap is used to capture cyclical conditions (see columns 1 to 4). By contrast, the debt ratio does not affect, or even increases, health expenditure in other OECD countries at high and low levels of debt, respectively. More importantly, the reaction of health expenditure to the debt ratio, observed in highly indebted EMU members, does not survive the introduction of the deficit ratio in the panel regression. As shown in columns 5 and 6, the deficit has a negative impact in all country groups except low-indebted EMU members, and this effect is significant when health expenditure is measured as a percentage of total expenditure. The last column of Table 7 shows that expenditure on public health stays constant or even declines as total expenditure increases.

Hence, contrary to the cases of gross capital formation and education expenditure, debt sustainability concerns appear not to affect the health spending decisions of highly indebted governments. The reduction of health expenditure seems instead related to the increase of other types of expenditures and the need to contain budget deficits. It is, however, difficult to explain this evidence with the constraints imposed by the SGP because the deficit reduces health expenditure, at high levels of debt, both inside and outside the euro area, but not in low-indebted EMU members.

Notes to Table 7

^aIn columns 3 and 4, 'HP output gap' is used as an explanatory variable instead of 'GDP growth rate'.

^bIn columns 7 and 8, 'Total government expenditure to GDP ratio' is used as an explanatory variable instead of 'Deficit to GDP ratio'.

Note: Robust standard errors are given in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

TABLE 7 Government expenditure on health: EMU countries versus other OECD countries

	(1) Health/Tot.exp	(2) Health/Y	(3) ^f Health/Tot.exp	(4) ^e Health/Y	(5) Health/Tot.exp	(6) Health/Y	(7) ^b Health/Tot.exp	(8) ^b Health/Y
Debt EMU high _{t-1}	-0.240** (0.099)	-0.111* (0.060)	-0.252** (0.109)	-0.155*** (0.053)	-0.143 (0.115)	-0.085 (0.056)	-0.000 (0.085)	-0.063 (0.063)
Debt EMU low _{t-1}	0.069 (0.059)	0.006 (0.012)	0.086 (0.084)	-0.002 (0.049)	0.073 (0.054)	0.012 (0.008)	0.073 (0.052)	0.004 (0.018)
Debt others high _{t-1}	-0.145 (0.188)	-0.001 (0.135)	-0.149 (0.192)	-0.017 (0.140)	0.006 (0.163)	0.016 (0.152)	0.067 (0.161)	0.052 (0.149)
Debt others low _{t-1}	0.273* (0.133)	0.349* (0.187)	0.265* (0.146)	0.357* (0.206)	0.282** (0.127)	0.372* (0.181)	0.310** (0.120)	0.305** (0.139)
GDP growth EMU high _{t-1}	0.007 (0.006)	-0.013*** (0.004)	-0.008 (0.006)	-0.013** (0.006)	-0.002 (0.005)	-0.016*** (0.004)	-0.012** (0.005)	-0.017*** (0.004)
GDP growth EMU low _{t-1}	-0.005* (0.002)	-0.013*** (0.004)	0.003 (0.006)	-0.004 (0.007)	-0.004** (0.002)	-0.012*** (0.003)	-0.007*** (0.002)	-0.013*** (0.003)
GDP growth others high _{t-1}	0.011*** (0.004)	-0.006 (0.005)	-0.006 (0.004)	-0.009* (0.005)	-0.007 (0.006)	-0.010 (0.006)	-0.007 (0.009)	-0.012 (0.008)
GDP growth others low _{t-1}	0.009 (0.008)	0.001 (0.006)	-0.000 (0.008)	0.009 (0.010)	0.005 (0.007)	0.001 (0.006)	0.007 (0.005)	0.006 (0.005)
Deficit EMU high _{t-1}					-0.019*** (0.002)	-0.005* (0.003)	-1.179*** (0.157)	-0.248 (0.151)
Deficit EMU low _{t-1}					0.002 (0.002)	0.003* (0.002)	-0.264*** (0.081)	0.128 (0.117)
Deficit others high _{t-1}					-0.022*** (0.005)	-0.005 (0.005)	-1.236*** (0.191)	-0.399* (0.232)
Deficit others low _{t-1}					-0.014** (0.006)	-0.004 (0.007)	-0.278 (0.335)	0.576* (0.291)
Constant	2.808*** (0.276)	1.638*** (0.250)	2.853*** (0.295)	1.699*** (0.262)	2.553*** (0.278)	1.569*** (0.245)	5.398*** (0.476)	1.595*** (0.465)
No. of observations	346	347	347	348	341	341	341	341
R ²	0.228	0.263	0.213	0.250	0.383	0.306	0.412	0.338
No. of countries	26	26	26	26	26	26	26	26

TABLE 8 Government expenditure on health: EU-15 countries versus new member states

	(1) Health/Tot.exp	(2) Health/Y	(3) ^a Health/Tot.exp	(4) ^a Health/Y	(5) Health/Tot.exp	(6) Health/Y	(7) ^b Health/Tot.exp	(8) ^b Health/Y
Debt EU-15 high _{t-1}	-0.252*** (0.084)	-0.112** (0.050)	-0.262** (0.092)	-0.149*** (0.044)	-0.146 (0.102)	-0.089* (0.047)	-0.025 (0.081)	-0.073 (0.055)
Debt EU-15 low _{t-1}	0.100* (0.053)	0.036 (0.026)	0.111 (0.068)	0.005 (0.034)	0.100 (0.059)	0.036 (0.023)	0.107** (0.051)	0.024 (0.023)
Debt NMS low _{t-1}	-0.040 (0.093)	0.040 (0.126)	-0.033 (0.260)	0.086 (0.211)	0.023 (0.092)	0.062 (0.127)	0.044 (0.107)	0.031 (0.120)
GDP growth EU-15 high _{t-1}	0.006 (0.006)	-0.012*** (0.003)	-0.007 (0.005)	-0.012** (0.005)	-0.002 (0.004)	-0.014*** (0.004)	-0.009* (0.004)	-0.015*** (0.004)
GDP growth EU-15 low _{t-1}	-0.000 (0.005)	-0.010*** (0.005)	0.003 (0.004)	-0.008* (0.005)	-0.002 (0.003)	-0.010** (0.004)	-0.002 (0.005)	-0.007 (0.006)
GDP growth NMS low _{t-1}	0.036*** (0.009)	0.022*** (0.005)	-0.016 (0.026)	0.012 (0.008)	0.025** (0.009)	0.018*** (0.005)	0.027*** (0.006)	0.023*** (0.005)
Deficit EU-15 high _{t-1}					-0.018*** (0.003)	-0.004 (0.002)	-1.110*** (0.156)	-0.200 (0.140)
Deficit EU-15 low _{t-1}					-0.005 (0.005)	0.002 (0.001)	-0.209*** (0.065)	0.387* (0.208)
Deficit NMS low _{t-1}					-0.023*** (0.002)	-0.008*** (0.002)	-0.833*** (0.064)	0.085* (0.049)
Constant	3.190*** (0.259)	2.084*** (0.160)	3.238*** (0.296)	2.168*** (0.162)	2.918*** (0.301)	2.021*** (0.152)	5.998*** (0.330)	2.176*** (0.373)
No. of observations	260	260	261	261	257	257	257	257
R ²	0.269	0.171	0.210	0.114	0.411	0.194	0.482	0.196
No. of countries	19	19	19	19	19	19	19	19

^aIn columns 3 and 4, 'HP output gap' is used as an explanatory variable instead of 'GDP growth rate'.

^bIn columns 7 and 8, 'Total government expenditure to GDP ratio' is used as an explanatory variable instead of 'Deficit to GDP ratio'.

Note: Robust standard errors are given in parentheses. *** p < 0.01; ** p < 0.05; * p < 0.1.

To investigate the relation between health expenditure and the debt ratio in NMS countries, in Table 8 we compare the behaviour of low-debt NMS countries (thus excluding Hungary) with those of high-debt EU-15 countries and of low-debt EU-15 countries.

Evidence for EU member states is consistent with our findings in Table 7. In low-debt EU-15 and NMS countries, the debt ratio has no effect on health expenditure, while in highly indebted countries the effect largely vanishes after controlling for the deficit ratio (columns 5 and 6). More importantly, we find evidence that health expenditure is reduced as the deficit increases, both in highly indebted EU-15 countries and in low-debt NMS countries where the effect is significant no matter how health expenditure is normalised. Indeed, as for education expenditure, it appears that NMS governments have cut expenditure on public health as total expenditure and the deficit increased, even at low levels of debt.

The finding of a similar negative relation between health expenditure and the government deficit at high levels of debt in both EMU member states and other OECD countries and the absence of such a relation in low-debt EMU and EU-15 members may suggest that the formal constraints of the SGP have not made a substantial difference. However, there is some evidence that problems in controlling total expenditure, high deficits and the rules of the preventive arm of the SGP may have distorted expenditure allocation in NMS countries.

V. Conclusions

We have documented that, at high levels of debt, governments cut capital expenditure, as the debt ratio increases, in all OECD countries independently of EMU (or EU) membership. Government expenditure on education is also reduced relative to other expenditures in highly indebted OECD countries as the debt ratio increases. The effect is as strong outside the euro area as in EMU countries. By contrast, neither capital expenditure nor education expenditure is affected by the debt ratio in low-debt countries.

The SGP does not appear to set an additional constraint on the fiscal policies of EMU (or EU-15) member states in that capital expenditure and education expenditure are significantly reduced in all highly indebted OECD countries. Furthermore, the negative reaction of capital and education expenditures to the debt ratio is robust to the inclusion of the government deficit in the panel regressions.

Our findings are consistent with the evidence on capital expenditure provided by Galí and Perotti (2003), who argue against the relevance of the constraints imposed by the SGP, but qualify their results in an important way: the decline of public investment observed in all OECD countries is due not just to historical factors but to a common concern for debt sustainability.

The same concern negatively affects education expenditure, whereas health expenditure appears to be unrelated to the debt ratio.

At high levels of debt, debt sustainability imposes a constraint on fiscal policy that is strong enough to distort the allocation of government expenditure even in the absence of formal rules such as the SGP. We have shown that this behaviour can be easily rationalised in a simple growth model where the government must cut public investment in order to ensure debt sustainability as the gross debt reaches a threshold level.

Fixed capital formation in low-debt new member states also appears not to be affected by the rules of the SGP. In these countries, public investment even increases with the debt ratio, a finding that is reassuring for their growth prospects. A different picture, however, emerges when we focus on expenditures on public health and education, as it appears that NMS governments cut such expenditures – even at low levels of debt – as the deficit increases. Problems in controlling total expenditure, high deficits and the rules of the preventive arm of the SGP may have penalised investment in human capital while leaving fixed capital investment unaffected.

Our analysis points to important distortionary effects of fiscal restraint on the allocation of public funds that penalise fixed capital and human capital formation at high levels of debt. Indeed, debt sustainability concerns (or high deficits in the case of the NMS) may lead to significant composition effects on public expenditure. Such effects appear substantial in the case of the NMS, where the shares of education and health expenditure fall with an increase in government deficit, while capital expenditure does not suffer from high deficits. A tentative explanation of this evidence is that investment in fixed capital is favoured over investment in human capital, say in education, because of its greater reversibility. Alternatively, human capital income could be perceived as a less reliable source of revenues when migration opportunities make labour income a much more elastic tax base than capital income.

While the effects of fiscal consolidation on government capital expenditure have been studied widely, there has so far been scant attention given to the allocation of public resources between investment in fixed capital and in human capital and across different types of investment. Though preliminary, the evidence that we provide suggests that composition effects are substantial and that understanding their determinants is an important task for future research.

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